Assignment 3

Persistent and Non-Persistent HTTP Connections

Name: Arnav Singhal
MIS: 612203012
TY B. Tech. Comp. Div 1
Batch: T3

<u>Problem Statement:</u>
Create a web page with N (e.g. 10) embedded images.
Configure your browser (Firefox) with following settings (each setting requires repeat of experiment) ·
Non persistent connection and persistent connections.
Note down the time taken to display the entire page in each of the settings. Ensure that (cache is cleared before starting the web request). Explain the response time differences. What is the optimal number of persistent connections for best performance? Explain your answer.
Use Wireshark to capture the packets between the client and the server while the file is accessed.
Calculate the time taken to load objects from the server for non-persistent and persistent connections.
Find out the optimal number of HTTP persistent connections based on your observations.

Setup

- Server Side: A Python-based HTTP server hosted on the local loopback address (127.0.0.1) listening on Port 8000. The server was initialized using the command python3 -m http.server 8000
- Client Side: Mozilla Firefox Version 110.0
- Network Capture: Wireshark capturing on the Loopback: lo interface, filtered for tcp.port == 8000 to isolate experiment traffic from background system noise.

Webpage used (index.html)

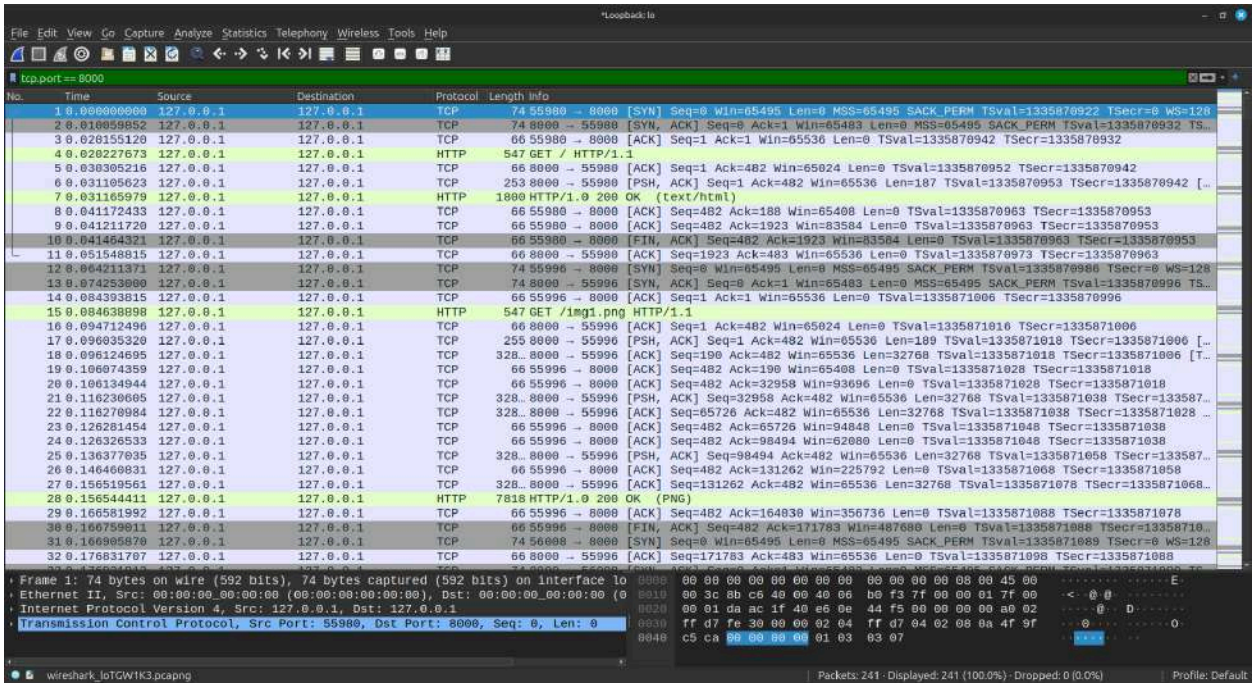
```
<!DOCTYPE html>
<html>
	<head>
		<title>Linux Networking Commands</title>
	</head>
	<body>
		<h1>Linux Networking Commands</h1>
		<ol>
			<li>
				<h3>ifconfig</h3>
				<img src="img1.png" width="1366" height="768">
			</li>
			<li>
			<h3>ip</h3>
			<img src="img2.png" width="1366" height="768">
		</li>
			<li>
			<h3>traceroute</h3>
			<img src="img3.png" width="1366" height="768">
		</li>
			<li>
			<h3>tracepath</h3>
			<img src="img4.png" width="1366" height="768">
		</li>
			<li>
			<h3>ping</h3>
			<img src="img5.png" width="1366" height="768">
		</li>
			<li>
			<h3>netstat</h3>
			<img src="img6.png" width="1366" height="768">
		</li>
			<li>
			<h3>nslookup</h3>
			<img src="img7.png" width="1366" height="768">
		</li>
			<li>
			<h3>dig</h3>
```

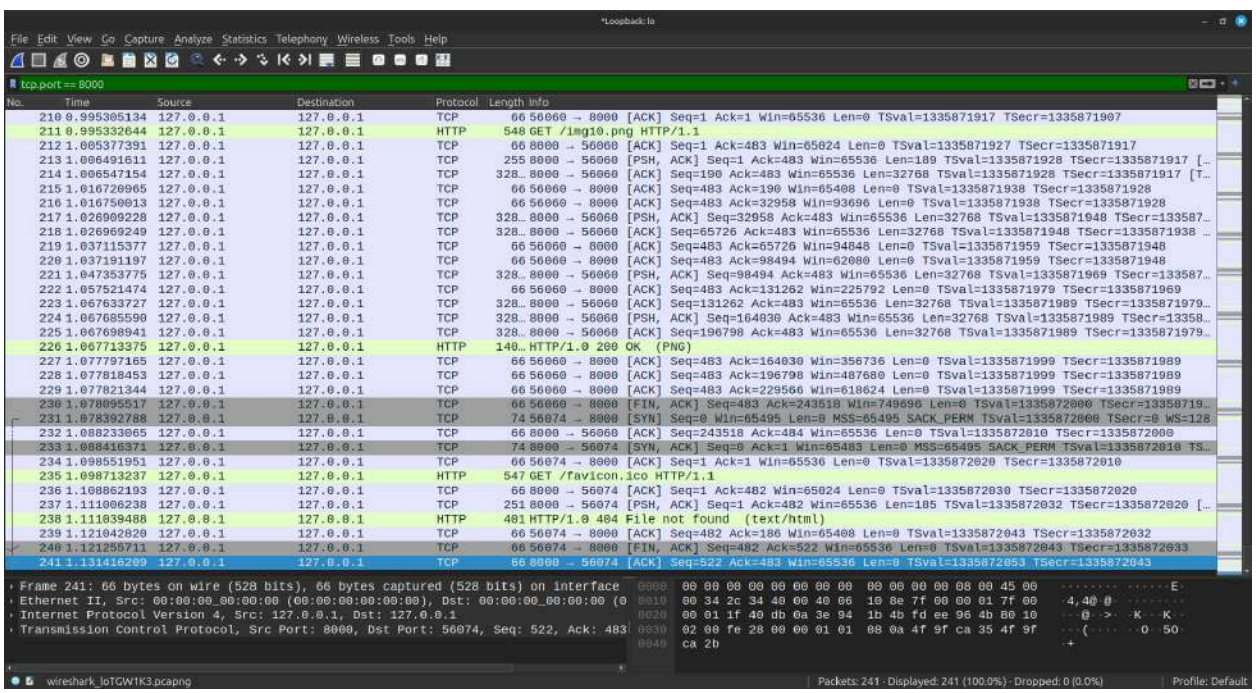
```
                <img src="img8.png" width="1366" height="768">
            </li>
                <li>
                <h3>route</h3>
                <img src="img9.png" width="1366" height="768">
            </li>
                <li>
                <h3>host</h3>
                <img src="img10.png" width="1366" height="768">
            </li>
           </ol>
    </body>
</html>
```

1. Non-Persistent HTTP Connections

<u>Browser Settings:</u> network.http.keep-alive set to false in browser configuration settings (about:config)

Observations:





Start Packet (1): 0.000000000 s
End Packet (243): 1.131416209 s
Time taken to display entire webpage: 1.131416209 s

Here, 11 separate TCP connections are used, one for each object (HTML file + 10 images). An additional connection is opened at the end for the webpage's favicon, but no object is transmitted over this connection.

2. Persistent HTTP Connections

<u>Browser Settings:</u>

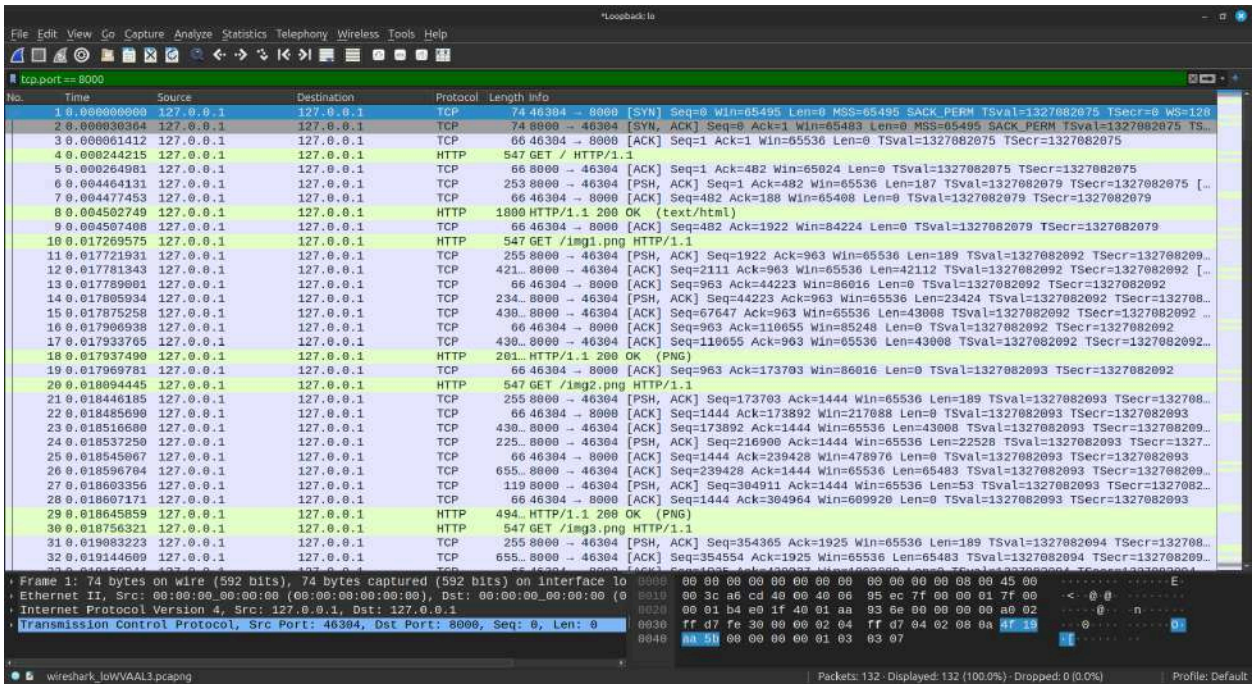
- network.http.keep-alive set to true in browser configuration settings (about:config)
- network.http.max-persistent-connections-per-server set to various values to determine the optimal number of persistent connections.

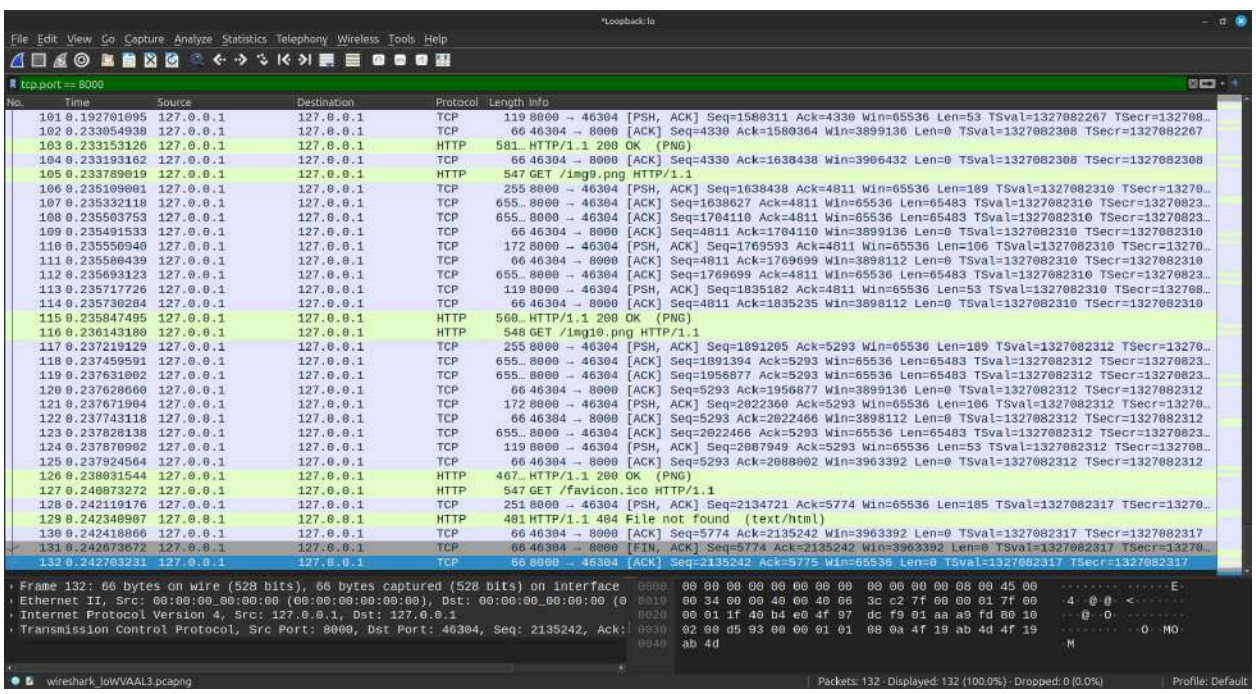
Unlike non-persistent HTTP connections, persistent connections allow transmission of more than one object over the same connection.

When only a single persistent connection is used, all objects are transmitted over it sequentially before the connection is closed. However, modern browsers may open multiple parallel persistent HTTP connections (upto a limit set in the browser's configuration settings). In such a case, varying numbers of objects will be transmitted over each of the connections parallelly.

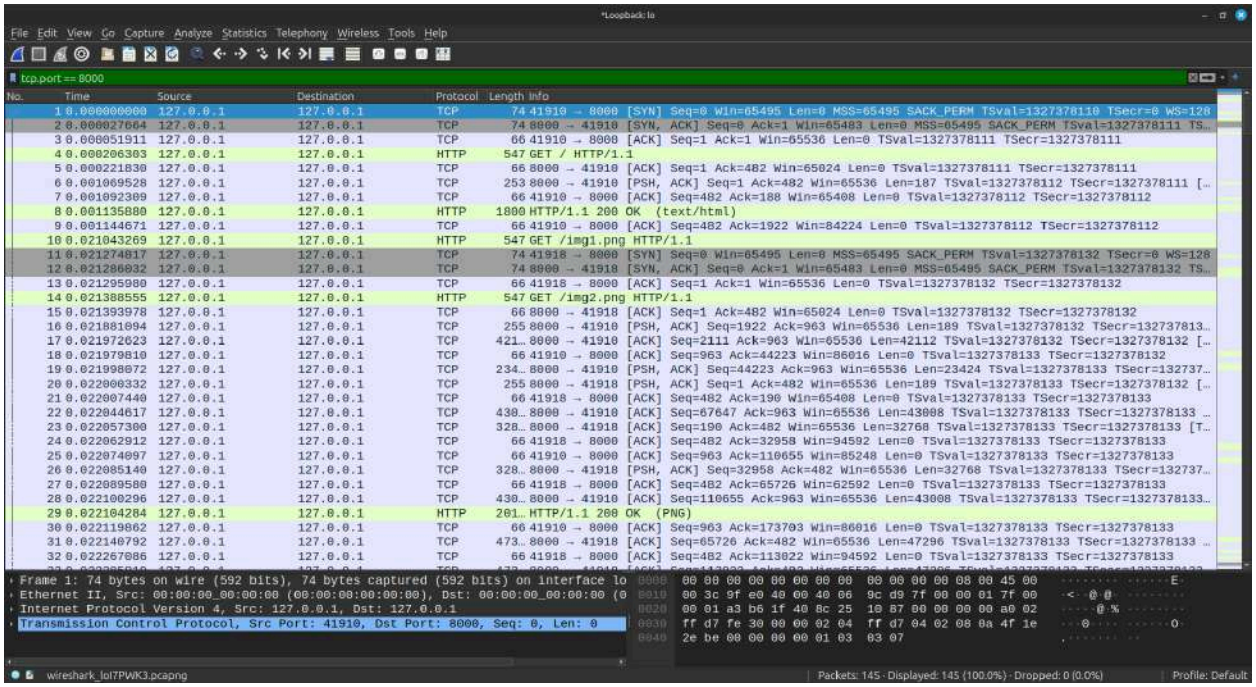
Observations:

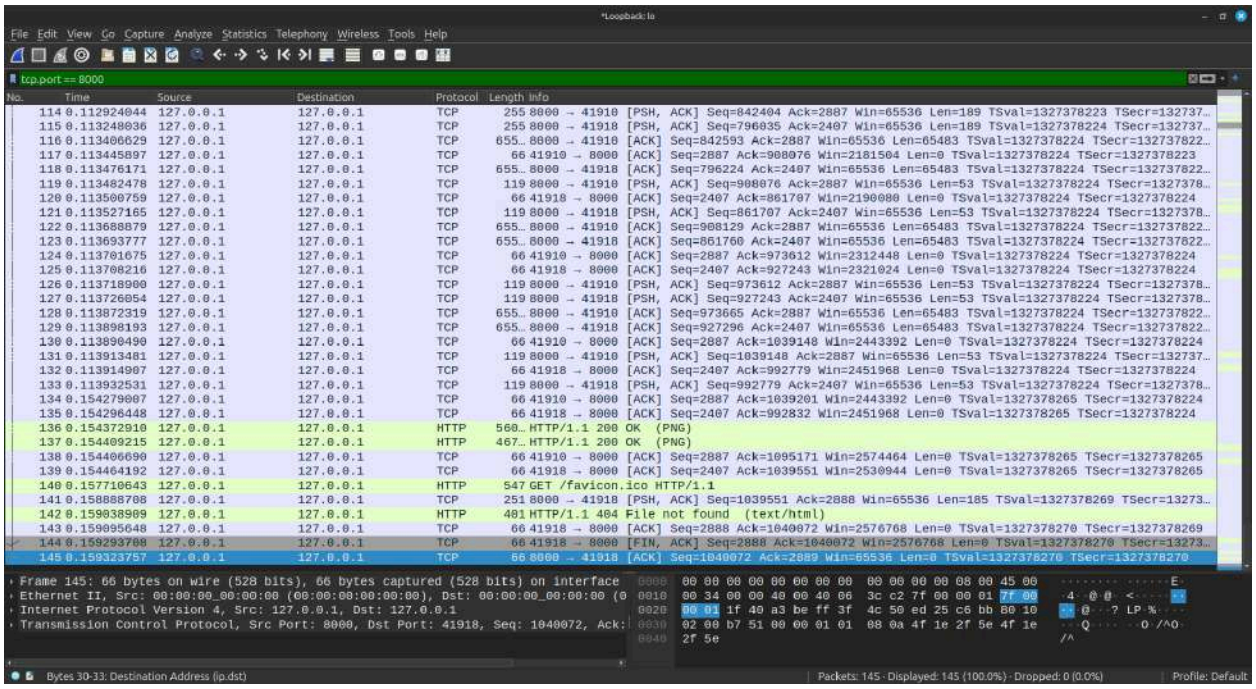
1) 1 persistent connection (Max: 1)



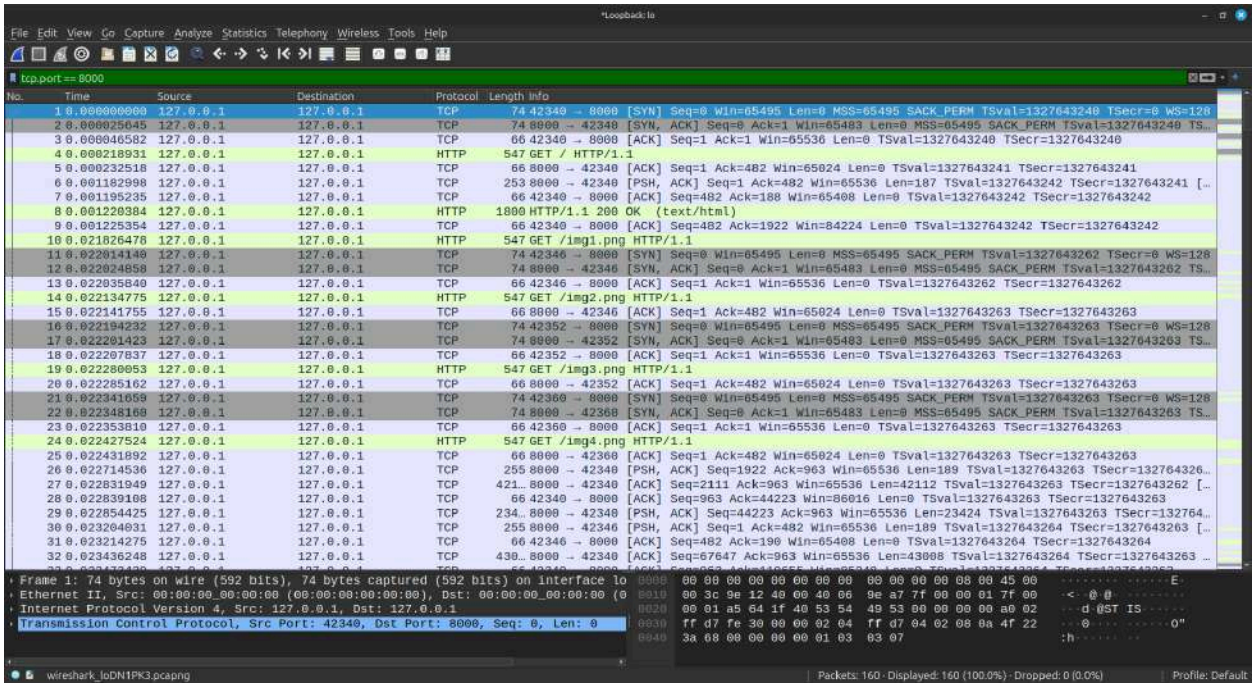


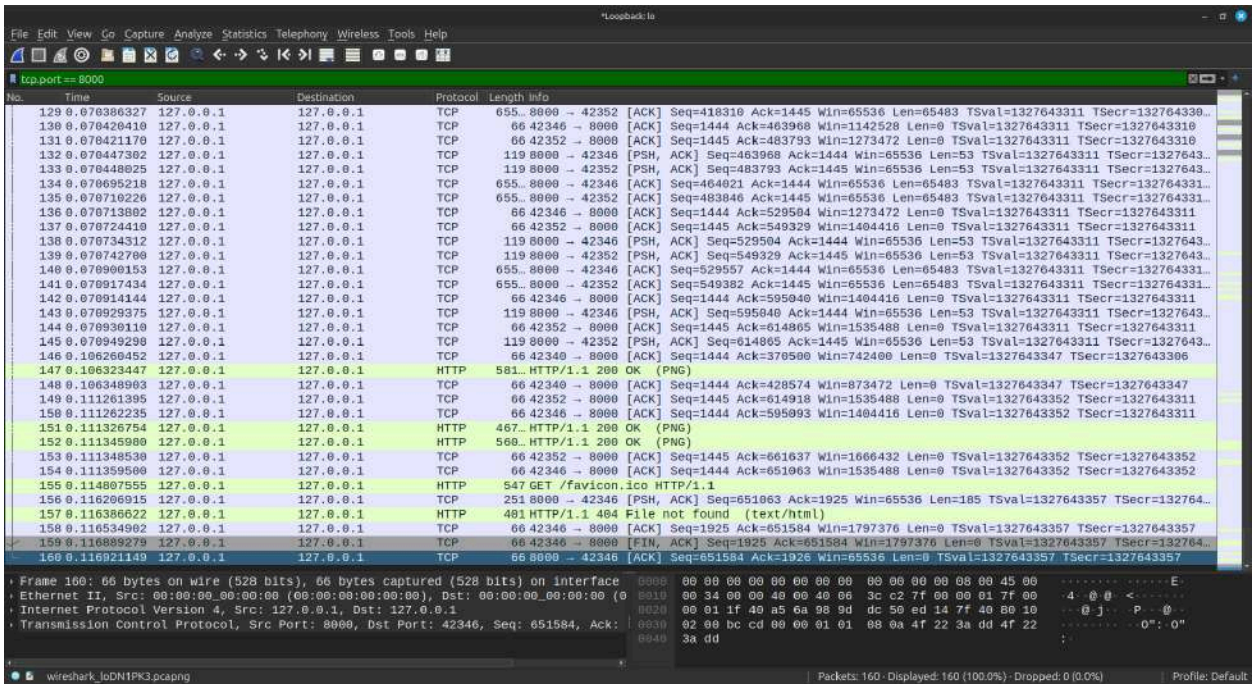
2) 2 persistent connections (Max: 2)



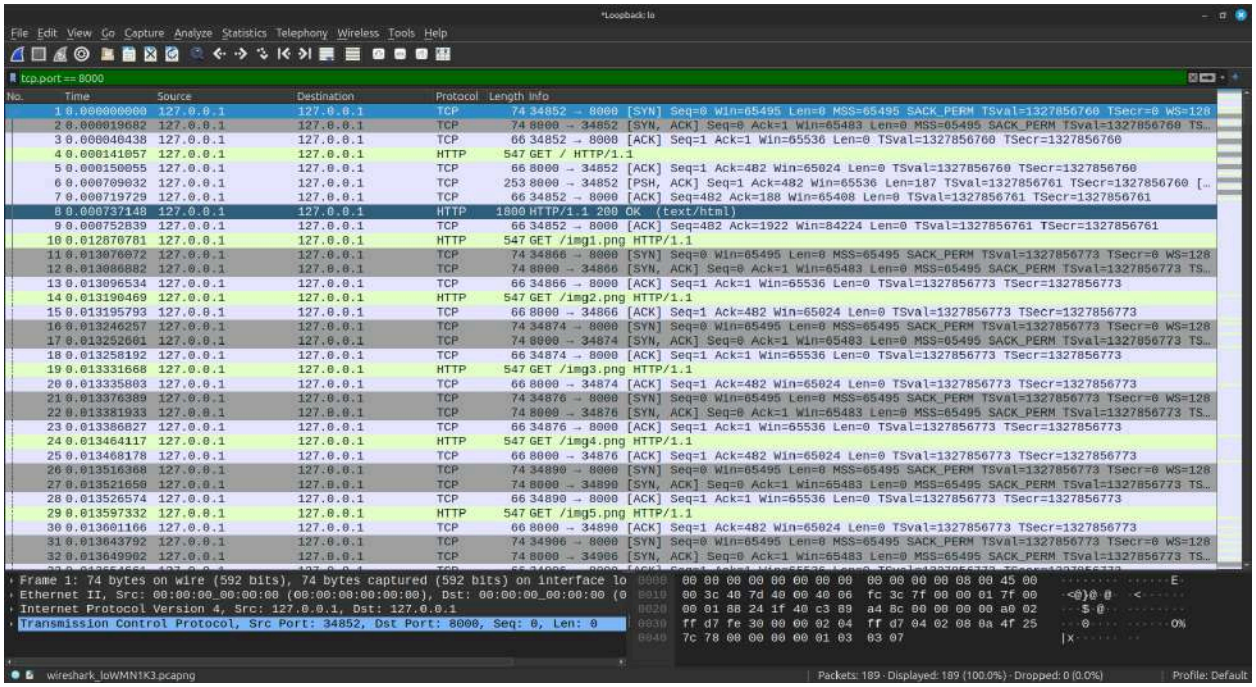


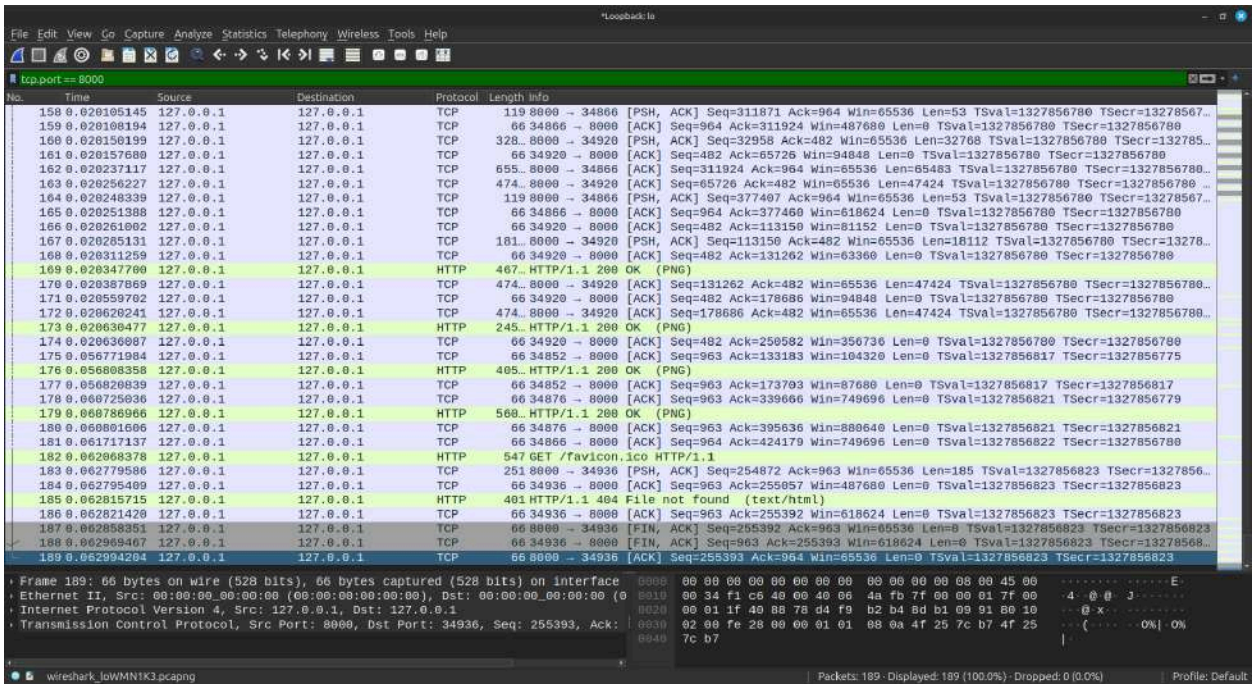
3) 4 persistent connections (Max: 4)



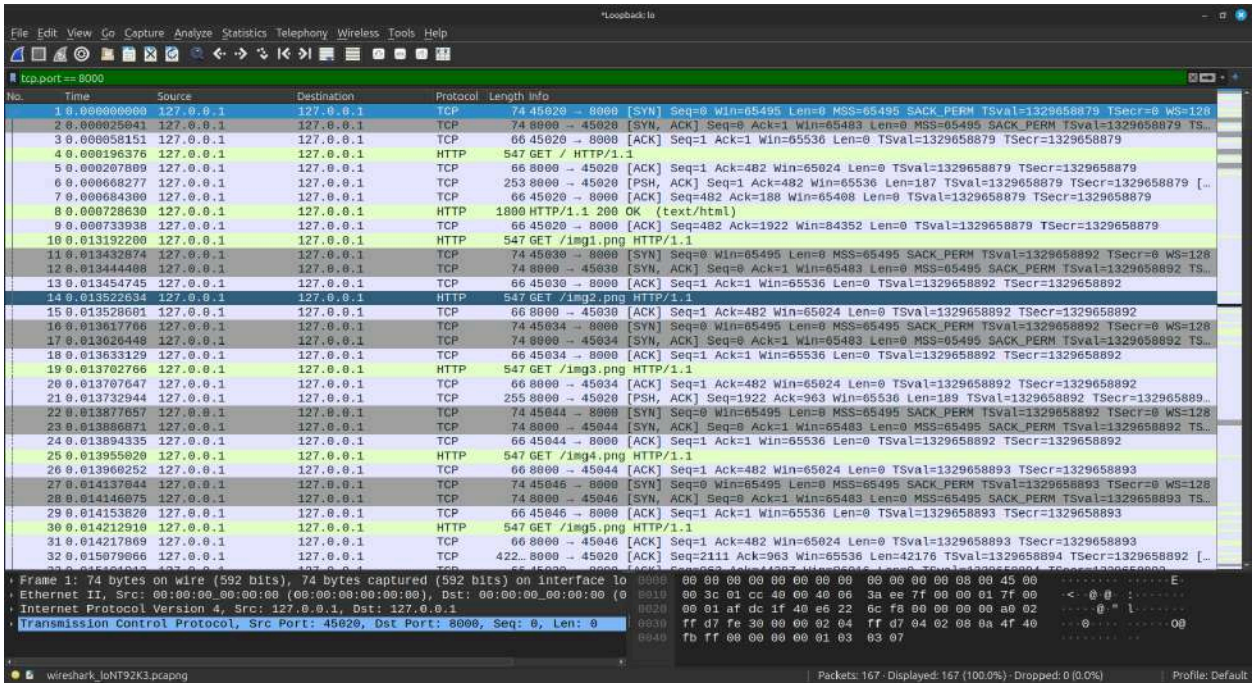


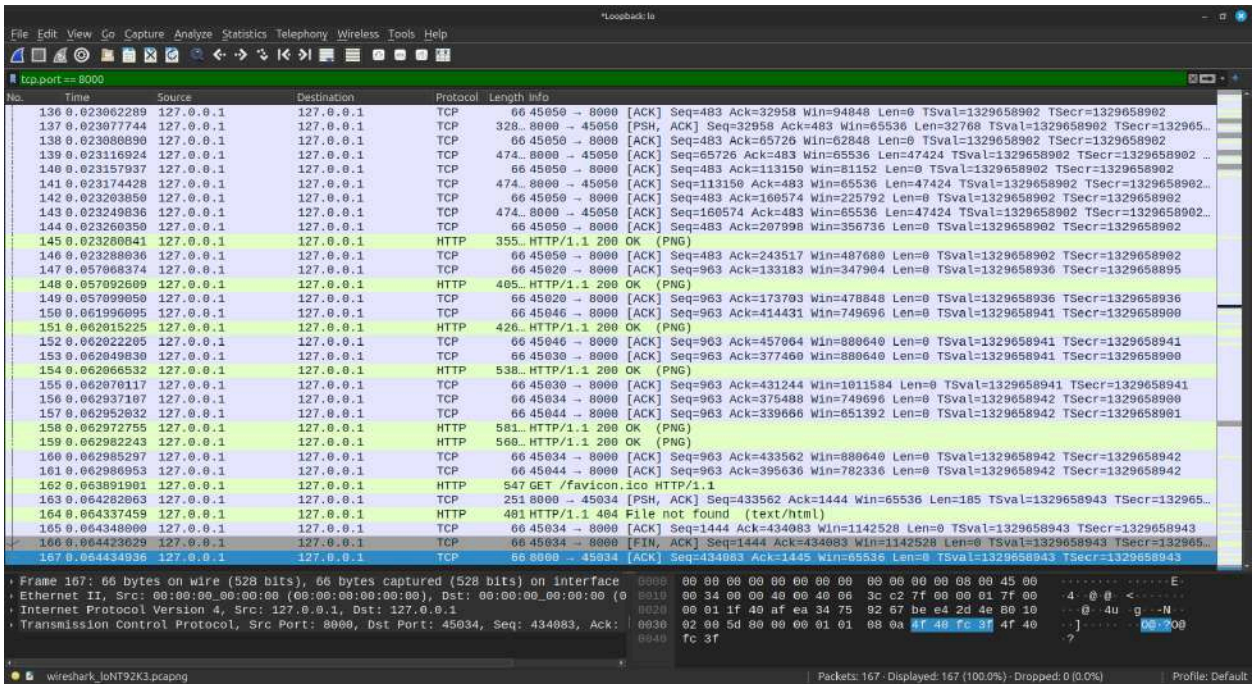
4) 7 persistent connections (Max: 8)



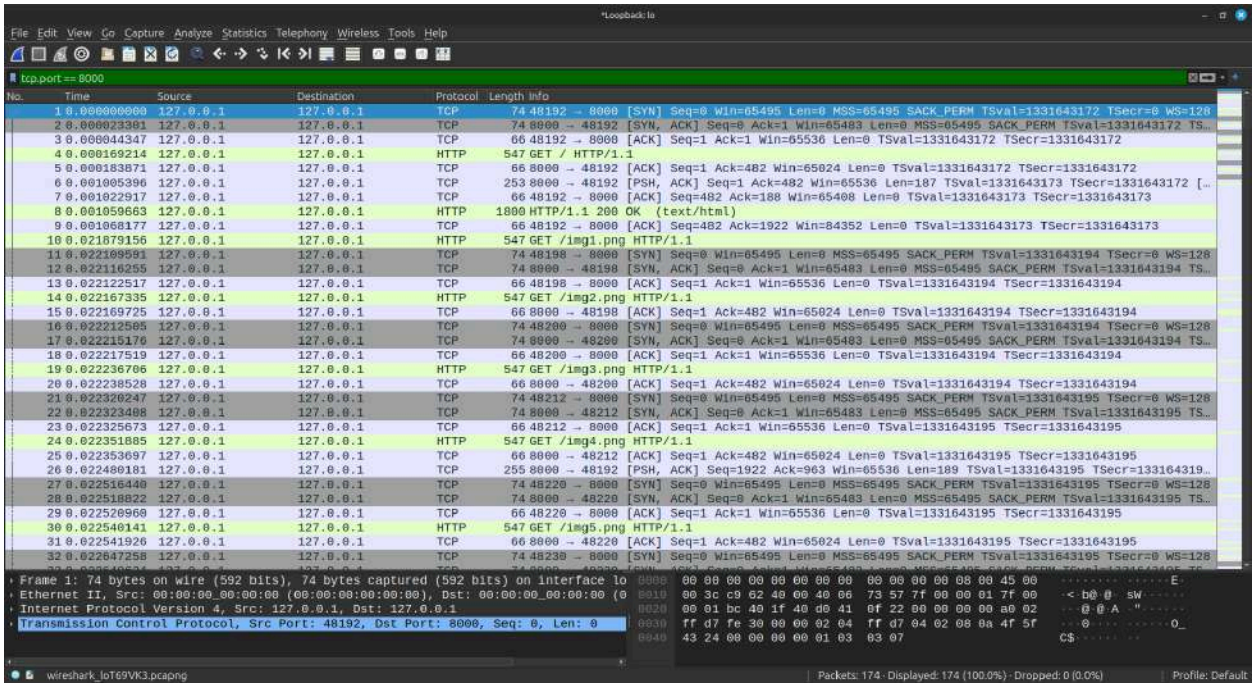


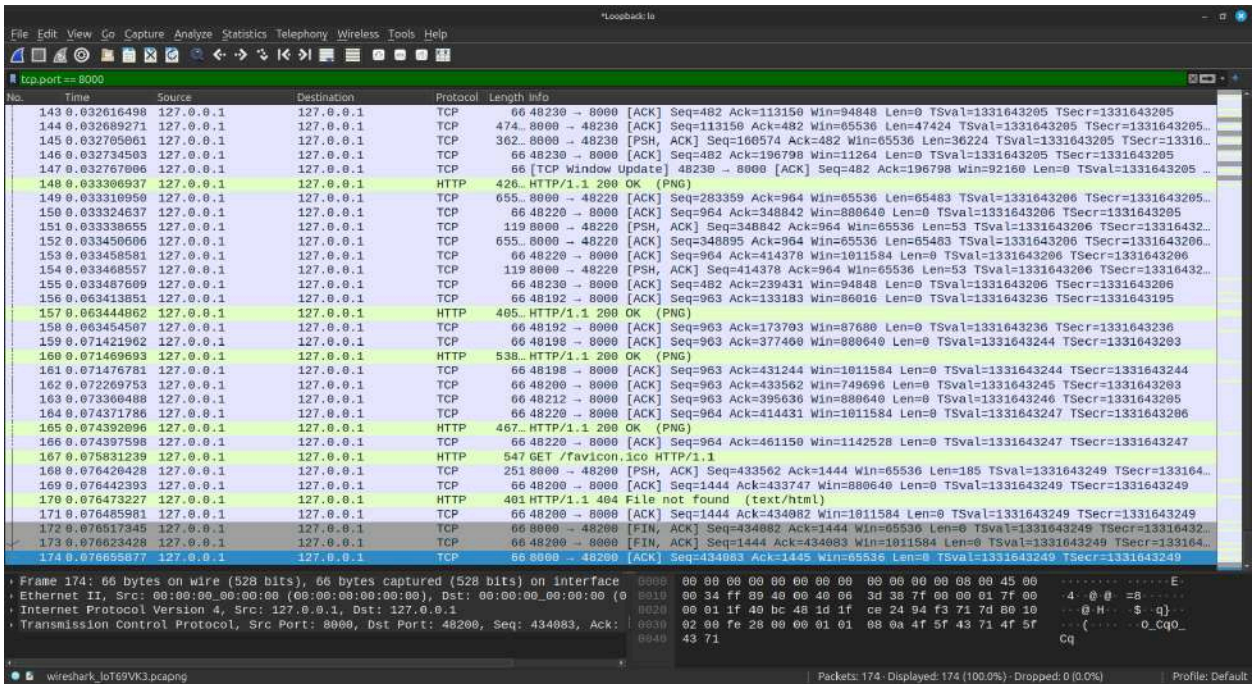
5) 8 persistent connections (Max: 8)





6) 6 persistent connections (Max: 6)





No. of persistent connections	Time taken to load webpage (in s)
1	0.242703231
2	0.159323757
4	0.116921149
6	0.076655877
7	0.062994204
8	0.064434936

Explanation for difference in response time

1. Persistent v/s Non-Persistent

The response time for persistent connections is substantially lower than that for non-persistent connections due to the following reasons:

- TCP Handshaking: When non-persistent connections are used, every object (including the HTML and all 10 images) incurs a full Round-Trip Time (RTT) delay for the TCP three-way handshake before data transfer can begin.
- OS Overhead: The non-persistent approach requires the operating system to constantly allocate and deallocate socket resources (ports), creating kernel-level scheduling delays that are absent in the streamlined persistent model.

2. Variation with number of non-persistent connections

For persistent connections, the response varies with the number of parallel connections used.

- Initially, increasing the connection limit (from 1 to 7 in this case) shows a steep improvement in performance by allowing multiple objects to be downloaded in parallel.
- However, beyond a certain limit (>7), performance does not improve and retransmissions are observed due to congestion and server limitations.

- Since the client opens many connections at the same time, the local TCP stack / server may be overloaded, and hence, the server cannot reply to every SYN request in the required amount of time.

Optimal number of persistent connections

From the observations, the optimal number of persistent connections is 7, as it provides the lowest page load time without causing congestion or retransmissions.

Conclusion

- The results show that persistent HTTP connections significantly improve webpage load time compared to non-persistent connections.
- Using two or more persistent connections in parallel provides a further improvement in performance upto a limit (in this case, 7). Increasing the number of parallel connections beyond this point leads to a drop in performance due to congestion and retransmissions.
- Hence, in general, a moderate number of persistent connections (4 - 8) gives the best performance.